

ABSTRACT OF THE DISSERTATION

A Predictive Model for Inpatient Major Joint Replacement or Reattachment of Lower Extremity

by FORREST PASCAL

Dissertation Director:
Shankar Srinivasan

Many have suggested that the United States healthcare system is broken. Costs are higher than ever, access is limited, and at times quality is questionable. One area of opportunity to lower cost, increase quality, and provide greater access is major joint replacement or reattachment of lower extremity—the most common inpatient surgical procedure for Medicare beneficiaries. Existing research points to emergency department visits, readmissions, and mortality as strong determinants of risk in an inpatient stay for major joint replacement or reattachment of lower extremity. For the current study, approximately 2.3 million inpatient claims were ingested from Medicare, resulting in 74 187 major joint replacement claims being extracted, cleansed, processed, and transformed. For each claim, emergency department visits, readmissions, mortality, and length of stay were calculated, along with the creation of an ICD crosswalk from 9 to 10 and Elixhauser Comorbidity Indexes for mortality & readmissions. A novel algorithm was developed to determine the risk of each claim. SAS Enterprise Miner, MATLAB, and MLJAR were used to mine the claims using supervised machine learning algorithms, and Tableau was used to visualize correlations and create 2D plots. This research provided the following insights: Matlab's Ensemble Boosted Tree algorithm

predicted the novel risk 8.out of 10 times across both the training and test dataset, proving its portability and reliability. Consistently, the physicians, provider (hospital), claim payment, type of admission and beneficiary county yielded the strongest predictor strength in predicting the outcome novel risk derived from emergency department visits, mortality, and readmissions. These predictors present areas of opportunity to lower cost, increase access, and improve quality by being used as indicators for early warning & surveillance systems for case workers, clinicians, and hospital administrators. Furthermore, machine learning models utilized in value-based care can assist healthcare leaders, payers, and providers with decision making on which care models may be most effective in facilitating associations to data on outcomes about patients with the highest risk profiles—specifically to identify which patients to follow more closely, which physicians and hospitals have the most successful results, and which geographic areas have differing results. Additionally, white females made up over 60% of observations, and both white females and males had the costliest claim payments. Lastly, obesity and hypertension (complicated & uncomplicated) were the most frequent comorbidities across gender, race and age group.